

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

Subject Code:BE01000121

Date:13-01-2025

Subject Name:Programming for Problem Solving

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Write an algorithm to check whether the entered number is Even or Odd.	03
	(b) Describe various symbols used for preparing flow chart.	04
	(c) Explain different type of operators used in c language with their precedence and associativity.	07
Q.2	(a) Differentiate: Exit controlled Loop and Entry controlled Loop	03
	(b) Explain: a) break statement b) continue statement	04
	(c) Write a C program to find factorial of a number.	07
	OR	
	(c) Construct a C program to swap the values of two integers with use of only two variables.	07
Q.3	(a) Demonstrate the use of conditional operator statement with an example.	03
	(b) Explain the structure of switch case statement.	04
	(c) Write a program to print the triangle shown below. 1 1 0 1 0 1 1 0 1 0 1 0 1 0 1	07
	OR	
Q.3	(a) Write a program to check whether entered character is vowel or not?	03
	(b) Explain getch(), getchar(), gets(), puts() .	04
	(c) Write a program to add all the elements of an array.	07
Q.4	(a) What is structure? Explain with example how to declare a structure and how to initialize it.	03
	(b) Explain following string manipulation function. strcat(), strcpy(), strcmp() and strlen()	04

- (c) Define a Structures which contains details of a Cricketer: 07
- Name of Player:
 - Team name:
 - Total run scored:
 - Batting average:

OR

- Q.4** (a) Debug the following codes and show the output for this: 03
- ```
For (i=0 ; i<=50 ; i++);
{
printf("\n %d" , i)
}
```
- \*consider header files and declaration of variable is done.
- (b) Summarize the methods for initialization of One-Dimensional array. 04
- (c) Construct a C program to add 3X3 matrix. 07
- Q.5** (a) Differentiate between structure and union. 03
- (b) Explain fopen() and its mode with example. 04
- (c) Briefly explain any two file handling functions with an example. 07

**OR**

- Q.5** (a) Write True / False against following statements: 03
- 1) Pointer is a variable which holds address of location where value of other variable is stored.
  - 2) Pointer holds the address in form of float or integer.
  - 3) The address of a variable can be stored in two or more pointers.
- (b) Write a C program to copy content of one file to other with the help of file handling functions. 04
- (c) Develop a program in C to check the entered number is prime or not by creating a user-defined function named check\_prime(). 07

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**GUJARAT TECHNOLOGICAL UNIVERSITY****BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:BE01000121****Date:12-06-2025****Subject Name:Programming for Problem Solving****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|            |                                                                                                                            | <b>Marks</b> |
|------------|----------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Q.1</b> | <b>(a)</b> Draw block diagram of computer and explain functionalities of various components of computer.                   | <b>03</b>    |
|            | <b>(b)</b> Brief about computer hardware and software.                                                                     | <b>04</b>    |
|            | <b>(c)</b> Write algorithm or flowchart for finding maximum out of three numbers entered by user.                          | <b>07</b>    |
| <b>Q.2</b> | <b>(a)</b> Explain the structure of for loop with proper example.                                                          | <b>03</b>    |
|            | <b>(b)</b> Explain nested if-else statement with suitable example.                                                         | <b>04</b>    |
|            | <b>(c)</b> Explain different bit-wise operators with the example.                                                          | <b>07</b>    |
|            | <b>OR</b>                                                                                                                  |              |
|            | <b>(c)</b> Explain different string functions with example.                                                                | <b>07</b>    |
| <b>Q.3</b> | <b>(a)</b> Discuss the need of break and continue statements with example.                                                 | <b>03</b>    |
|            | <b>(b)</b> Explain recursion with example.                                                                                 | <b>04</b>    |
|            | <b>(c)</b> Explain Declaration, initialization and Printing on screen of a String using suitable program.                  | <b>07</b>    |
|            | <b>OR</b>                                                                                                                  |              |
| <b>Q.3</b> | <b>(a)</b> Explain while loop and do-while loop with example                                                               | <b>03</b>    |
|            | <b>(b)</b> What are header files? List any three with its usage.                                                           | <b>04</b>    |
|            | <b>(c)</b> Explain one dimensional array, two-dimensional array and multi-dimensional array with their syntax and example. | <b>07</b>    |
| <b>Q.4</b> | <b>(a)</b> What is pointer? Which arithmetic operations are valid on pointers?                                             | <b>03</b>    |
|            | <b>(b)</b> What is structure? How it is different from union? Explain nested structure with example.                       | <b>04</b>    |
|            | <b>(c)</b> Write a C program to find the largest number amongst three numbers entered by user.                             | <b>07</b>    |
|            | <b>OR</b>                                                                                                                  |              |
| <b>Q.4</b> | <b>(a)</b> What is pointer to pointer? Write suitable example to demonstrate the concept.                                  | <b>03</b>    |
|            | <b>(b)</b> Define Union in 'C' with suitable example.                                                                      | <b>04</b>    |
|            | <b>(c)</b> Write a C program to check if entered integer is a prime number or not.                                         | <b>07</b>    |
| <b>Q.5</b> | <b>(a)</b> Explain various file management function in C.                                                                  | <b>03</b>    |
|            | <b>(b)</b> Show the use of Malloc, Calloc, Free and Realloc in context of Dynamic memory allocation.                       | <b>04</b>    |
|            | <b>(c)</b> Explain error handling in file system with example.                                                             | <b>07</b>    |

**OR**

- Q.5**
- |            |                                                                                                                                         |           |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>(a)</b> | Explain debugging techniques.                                                                                                           | <b>03</b> |
| <b>(b)</b> | Differentiate call by value and call by reference.                                                                                      | <b>04</b> |
| <b>(c)</b> | Explain the following File Handling functions: 1. fseek( ) 2. ftell( ) 3. fread( ) 4. fwrite( ) 5. fscanf( ) 6. fprintf( ) 7. rewind( ) | <b>07</b> |

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Enrollment No./Seat No.:

## GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000121/BE01R00121

Date: 31-12-2025

Subject Name: Programming for Problem Solving

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

### Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

|                                                                                                                                        | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Q.1 (a)</b> Define followings: Compilers, Interpreters, Assemblers                                                                  | 03    |
| <b>(b)</b> Explain computer block diagram with functionalities of different components.                                                | 04    |
| <b>(c)</b> List different symbols used in flowchart with their functionality. Draw flowchart to find that given number is odd or even. | 07    |
| <b>Q.2 (a)</b> Write rules to declare variables in C language.                                                                         | 03    |
| <b>(b)</b> List and explain different primitive data types in C language.                                                              | 04    |
| <b>(c)</b> List and explain different operators in C language.                                                                         | 07    |
| <b>OR</b>                                                                                                                              |       |
| <b>(c)</b> Write C language program to find minimum and maximum from three numbers.                                                    | 07    |
| <b>Q.3 (a)</b> Explain switch case statement.                                                                                          | 03    |
| <b>(b)</b> Differentiate while loop and do-while loop.                                                                                 | 04    |
| <b>(c)</b> Write C language program to make sum of digits of given number. (hint: if number is 482 then sum is 4+8+2=14).              | 07    |
| <b>OR</b>                                                                                                                              |       |
| <b>(a)</b> Explain if-else statement.                                                                                                  | 03    |
| <b>(b)</b> Differentiate break and continue statement with example.                                                                    | 04    |
| <b>(c)</b> Write C language program to generate following triangle:<br>9<br>8 7<br>6 5 4<br>3 2 1 0                                    | 07    |
| <b>Q.4 (a)</b> What is an array? Brief about 2-D array.                                                                                | 03    |
| <b>(b)</b> Explain functions: strcpy(), strcat(), scanf(), main()                                                                      | 04    |

- (c) Create structure named as Book\_details with variables book\_id, title, author, and price. Create array of structures with it. From the array of structure show the book details having maximum price. 07

**OR**

- (a) What is union? How it is different from structure. 03
- (b) Discuss the parameter pass by value and parameter pass by reference with example. 04
- (c) Write C language program to check whether the given string is palindrome or not without using string functions. 07

- Q.5** (a) Explain debugging techniques. 03
- (b) Briefly explain fopen() function with different operating modes. 04
- (c) Write C language program to find factorial of a number using recursion. 07

**OR**

- (a) Explain working of pointer with an example. 03
- (b) Briefly explain memory allocation functions malloc() and calloc() with an example. 04
- (c) Write C language program to generate Fibonacci series using recursion. 07

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